

CSE 460: SOFTWARE ANALYSIS AND DESIGN

Homework #2 — Heart Health Imaging and Recording System

Fahad Alsulaiman

Item 1 Working Code

The complete implementation of the Heart Health Imaging and Recording System is written in Java with a JavaFX GUI. The source code is hosted on GitHub. The repository contains the full project including all source files, the build configuration, and sample data files demonstrating each of the four functional areas:

- **Patient Intake** — register new patients and schedule CT scan appointments
- **CT Scan Technician** — enter and save Agatston CAC score results
- **Patient View** — patients log in with their 5-digit ID to view CT results
- **Doctor View** — heart specialist reviews results, determines risk, and emails patient

GitHub repository link (code):

PLACEHOLDER_GITHUB_REPO_LINK

Item 2 Demo Video

A video demonstration (under 5 minutes) is provided showing the full system in action. The demo covers:

- a. How to run the project and login
- b. Patient Intake functionality — entering patient data and scheduling a scan
- c. CT Scan Tech functionality — recording Agatston CAC scores
- d. Patient View functionality — logging in by patient ID and viewing results
- e. Doctor View functionality — reviewing results, risk assessment, and sending the patient email

Demo video link (YouTube / Google Drive):

PLACEHOLDER_DEMO_VIDEO_LINK

Item 3 Final Implementation Class Diagram

The final class diagram reflecting the completed implementation is presented on the following page. It was produced in **Astah** and captures all packages, classes, fields, methods, and relationships of the delivered system.



